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## **ВЫПУСКНАЯ КВАЛИФИКАЦИОННАЯ РАБОТА**

### **MASTER THESIS**

*Тема Алгоритмы обнаружения и предотвращения каскадов мнений в реалистичных симуляциях социальных сетей с агентами на основе больших языковых моделей*

*Title Algorithms for Detecting and Mitigating Opinion Cascades in Realistic Social Network Simulations with LLM-Based Agents*

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# Abstract

This thesis develops a scalable method for eliminating majority illusion in a real social network graph and evaluates its behavioral implications in an LLM-based cascade simulation. Majority illusion occurs when a globally minority state appears locally dominant because of graph structure, especially in the presence of degree heterogeneity. Building on prior network-science and algorithmic work, the thesis formulates exact Majority Illusion Elimination (MIE) as an edge-editing optimization problem and then scales it by detecting communities and applying exact local treatment within each community. Under the simplifying assumption of equally sized communities, the resulting asymptotic cost decreases from  $O(n^4)$  for a monolithic exact treatment to  $O(n^4/C^3)$  for  $C$  communities. The implementation uses a public Instagram follower dataset as the structural substrate, converts users into LLM-based agents, generates ranked feeds, and simulates post creation, exposure, opinion updating, and follow / unfollow adaptation. Community-level experiments show that majority illusion can be removed with a small number of edge modifications. Baseline cascade experiments without intervention show non-trivial opinion dynamics and substantial computational cost. The thesis provides a unified framework that links formal graph intervention, scalable implementation, and agent-based behavioral validation.

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# Chapter 1

## Introduction

### 1.1 Motivation

Social platforms expose users to a highly incomplete view of the underlying social graph. In practice, users rarely observe the network as a whole; they interact with a ranked stream of posts, comments, recommendations, and social signals. What does a user actually see? Not the global distribution of opinions, but a small and algorithmically filtered local environment.

This gap between global structure and local perception makes network-level distortions especially important. One of the clearest examples is the *majority illusion*, introduced by Lerman et al. (2016). The phenomenon occurs when a globally minority attribute appears common in local neighbourhoods because high-degree nodes are disproportionately represented in the observations of others. In social media settings, perceived popularity affects attention, confidence, conformity, and sometimes behaviour itself.

The importance of majority illusion is threefold. First, it may bias subjective beliefs about what other people think, which is directly relevant to preference falsification, conformity, and social reinforcement (Granovetter, 1978; Jackson, 2010). Second, when adoption depends on observed behaviour in the local neighbourhood, distorted perception can facilitate opinion cascades even when the triggering opinion is globally rare (Centola, 2010). Third, modern platforms do not expose users to neighbourhoods uniformly. They rank, filter, and recommend content. This is crucial: structural distortions and ranking mechanisms may amplify one another.

This thesis studies how majority illusion can be detected, eliminated, and tested in a realistic simulation environment. More specifically, it asks how majority illusion elimination should be defined when opinions are fixed and the intervention acts on edges rather than labels; how such algorithms can be scaled to a real follower network with tens of thousands of users; and whether the resulting intervention changes cascade dynamics when users are represented by LLM-based agents.

The operational goal of the thesis is therefore to develop a scalable majority illusion elimination pipeline for real social network graphs and to validate it in a simulation with LLM agents. The pipeline combines exact local graph intervention with community decomposition. The graph is first partitioned into communities, then majority illusion is eliminated inside each community, and finally the treated graph is compared with the original graph in an agent-based cascade simulation.

## 1.2 Related Work

The conceptual starting point is the original majority illusion paper by Lerman et al. (2016), which explains how degree heterogeneity and degree–attribute correlations can make rare states appear common in local neighbourhoods. Their work connects majority illusion to the broader family of network paradoxes, including the friendship paradox, where local observations systematically differ from global averages.

The algorithmic study of majority illusion was later developed by Grandi et al. (2023), who initiated the computational analysis of identifying and eliminating majority illusion in social networks. Their results show that several related variants are computationally difficult and that structural restrictions are often needed to obtain tractable algorithms. This is important for the present thesis because it separates the intuitive idea of “fixing” a network from the actual algorithmic cost of doing so.

A particularly relevant result is due to Dippel et al. (2025), who show that complete majority illusion elimination can be solved in polynomial time by altering as few connections as possible. Their work considers three variants: edge addition, edge removal, and the combined Majority Illusion Elimination problem. The result is somewhat surprising because partially eliminating majority illusion is hard, while complete elimination admits polynomial-time algorithms. The present thesis builds on this edge-editing viewpoint, but adapts it to a practical setting where the graph is too large to treat as one monolithic optimization problem.

The broader diffusion literature explains why local perception matters for behaviour change. Threshold models formalize the idea that adoption depends on how many neighbours have already adopted a behaviour or opinion (Granovetter, 1978). In an online field experiment, Centola (2010) showed that network structure affects the spread of behaviour, supporting the view that clustered exposure and social reinforcement can change diffusion outcomes. These works justify treating graph structure not merely as a descriptive object, but as an intervention target.

The simulation component is informed by recent work on LLM-based generative agents. Park et al. (2023) show that large language models, combined with memory and interaction mechanisms, can generate coherent multi-agent social behaviour. This thesis does not assume that LLM agents are perfect models of human users. Rather, they are

used as controllable cognitive agents whose posts, feeds, opinion updates, and network reactions can be observed under different structural interventions. This makes it possible to test not only whether majority illusion is removed mathematically, but also whether the intervention changes cascade dynamics in a more realistic environment.

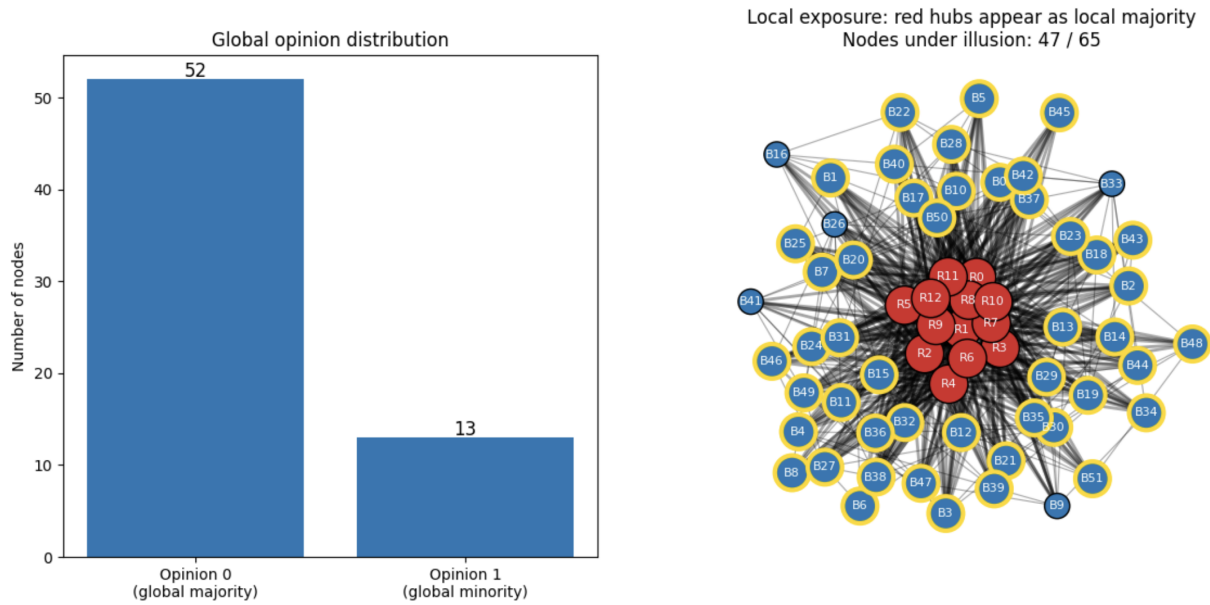


Figure 1.1: Illustrative network showing how a globally minority opinion can appear locally dominant because high-degree nodes are over-represented in neighbourhoods.

# Chapter 2

## Problem Formulation

### 2.1 Preliminaries

Let  $G = (V, E)$  be a simple graph, where  $V$  is the set of users and  $E$  is the set of social ties. For the analytical part of majority illusion elimination, the graph is treated as undirected. Each node  $v \in V$  is assigned a binary label

$$f(v) \in \{0, 1\}, \quad (2.1)$$

where, in the experiments reported here,  $f(v) = 1$  denotes the opinion whose apparent over-representation is undesirable at the neighbourhood level, while  $f(v) = 0$  denotes the reference or balancing opinion.

For a node  $v$ , let

$$N(v) = \{u \in V : \{u, v\} \in E\} \quad (2.2)$$

be its neighbourhood and let  $d(v) = |N(v)|$  be its degree. Define the counts

$$r(v) = \sum_{u \in N(v)} f(u), \quad b(v) = d(v) - r(v). \quad (2.3)$$

Thus  $r(v)$  is the number of neighbours with label 1 and  $b(v)$  is the number of neighbours with label 0.

**Definition 2.1** (Majority illusion). *Assume that label 1 is globally a minority state, i.e.*

$$\frac{1}{|V|} \sum_{v \in V} f(v) < \frac{1}{2}. \quad (2.4)$$

*A node  $v \in V$  is said to be under majority illusion if the neighbourhood majority disagrees with the global majority, namely if*

$$r(v) > b(v). \quad (2.5)$$

## 2.2 Majority Illusion Elimination

The considered intervention modifies the graph rather than the node labels. Let  $A$  be a set of added edges and  $R$  be a set of removed edges. The modified graph is

$$G' = (V, E'), \quad E' = (E \cup A) \setminus R. \quad (2.6)$$

**Definition 2.2** (Majority Illusion Elimination). *Given a graph  $G = (V, E)$ , a fixed binary labeling  $f$ , and a budgetless minimization objective, the Majority Illusion Elimination problem is:*

$$\min_{A, R} |A| + |R| \quad (2.7)$$

$$s.t. \ r'(v) \leq b'(v), \quad \forall v \in V. \quad (2.8)$$

where  $r'(v)$  and  $b'(v)$  are computed in  $G'$  under the same fixed labeling  $f$ .

This thesis focuses on the complete-elimination case because it is both theoretically crisp and directly aligned with the exact polynomial solvability result of Dippel et al. (2025). In contrast, partial elimination of majority illusion has a substantially harder algorithmic profile (Grandi et al., 2023).

## 2.3 Community-Local Variant

In the full Instagram graph, intervention is not performed on the entire graph simultaneously. Instead, the vertex set is partitioned into communities

$$V^* = \bigcup_{c=1}^C V_c, \quad V_c \cap V_{c'} = \emptyset \text{ for } c \neq c'. \quad (2.9)$$

For each community  $G_c = (V_c, E_c)$ , a local MIE problem is solved independently using the labels induced by the global initialization. This produces a treated intra-community edge set  $E'_c$ . The global treated graph is then reconstructed as

$$E^{\text{treat}} = E_{\times} \cup \bigcup_{c=1}^C E'_c, \quad (2.10)$$

where  $E_{\times}$  is the set of all original inter-community edges among vertices in  $V^*$ . Thus local treatment does not alter cross-community structure in the current implementation.

# Chapter 3

## Exact Majority Illusion Elimination Algorithm

### 3.1 Theoretical Background

Dippel et al. (2025) show that complete majority illusion elimination can be solved in polynomial time through linear optimization over total-dual-integral systems related to  $b$ -matching polytopes. Their result is especially important because it distinguishes the complete-elimination case from partial-elimination variants studied by Grandi et al. (2023). In the present work, however, the implementation follows a practical exact ILP formulation over candidate edge additions and removals inside each community. This choice is straightforward to implement, fits naturally with community decomposition, and remains exact on the selected local subgraphs.

### 3.2 Candidate Sets and Structural Restrictions

Let  $G_c = (V_c, E_c)$  be one treated community. Following the structural logic emphasized by Dippel et al. (2025), candidate sets are filtered before optimization:

- edges between two label-1 vertices are never added, because such additions increase local exposure to the minority label without helping elimination;
- edges between two label-0 vertices are never removed, because such deletions weaken rather than strengthen neighbourhood balance.

Accordingly, let  $A_c \subseteq \binom{V_c}{2} \setminus E_c$  be the admissible set of candidate additions and let  $R_c \subseteq E_c$  be the admissible set of candidate removals.

For each node  $v \in V_c$ , define the base counts in the untreated community:

$$b_c(v) = |\{u \in N_c(v) : f(u) = 0\}|, \quad r_c(v) = |\{u \in N_c(v) : f(u) = 1\}|. \quad (3.1)$$

Partition the candidate add/remove neighbourhoods according to neighbour labels:

$$A_c^0(v) = \{u : \{u, v\} \in A_c, f(u) = 0\}, \quad A_c^1(v) = \{u : \{u, v\} \in A_c, f(u) = 1\}, \quad (3.2)$$

$$R_c^0(v) = \{u : \{u, v\} \in R_c, f(u) = 0\}, \quad R_c^1(v) = \{u : \{u, v\} \in R_c, f(u) = 1\}. \quad (3.3)$$

### 3.3 ILP Formulation

Introduce binary variables

$$x_e \in \{0, 1\} \quad \forall e \in A_c, \quad z_e \in \{0, 1\} \quad \forall e \in R_c, \quad (3.4)$$

where  $x_e = 1$  means that a candidate non-edge  $e$  is added and  $z_e = 1$  means that an existing edge  $e$  is removed.

The post-intervention counts for each node  $v$  are

$$\tilde{b}_c(v) = b_c(v) + \sum_{u \in A_c^0(v)} x_{\{u, v\}} - \sum_{u \in R_c^0(v)} z_{\{u, v\}}, \quad (3.5)$$

$$\tilde{r}_c(v) = r_c(v) + \sum_{u \in A_c^1(v)} x_{\{u, v\}} - \sum_{u \in R_c^1(v)} z_{\{u, v\}}. \quad (3.6)$$

The optimization problem is

$$\min_{x, z} \quad \sum_{e \in A_c} x_e + \sum_{e \in R_c} z_e \quad (3.7)$$

$$\text{s.t.} \quad \tilde{r}_c(v) \leq \tilde{b}_c(v), \quad \forall v \in V_c, \quad (3.8)$$

$$x_e \in \{0, 1\}, \quad \forall e \in A_c, \quad (3.9)$$

$$z_e \in \{0, 1\}, \quad \forall e \in R_c. \quad (3.10)$$

Constraint (3.8) enforces the absence of majority illusion after modification. The objective (3.7) minimizes the total number of edited edges. In the implementation, this ILP is solved exactly on each community selected for local treatment.

### 3.4 Interpretation

The ILP makes the intervention mechanism transparent. Because the treatment acts on graph structure rather than content, the method is best interpreted as a structural intervention on exposure opportunities. In the simulation stage, these graph edits are translated into feed changes: added edges create additional exposure, while removed edges suppress previously available exposure.

# Chapter 4

## Scalability via Community Decomposition

### 4.1 Why Exact Global Treatment is Infeasible

The exact complete-elimination result of Dippel et al. (2025) is polynomial, but polynomial does not imply practical tractability for a graph with millions of nodes. The theoretical formulation in the source paper relies on a separation step whose worst-case running time is reported as  $O(n^4)$ , which motivates the cost model adopted in this thesis for a monolithic exact treatment.

If the full graph has  $n$  vertices, then a direct exact intervention can be abstracted as

$$T_{\text{exact}}(n) = O(n^4). \quad (4.1)$$

For the Instagram-scale graph considered here, this is far too expensive for repeated experimentation, especially when treatment must be combined with agent-based cascade simulation.

### 4.2 Community Detection with Louvain

To reduce the intervention cost, the graph is first partitioned into communities using the Louvain algorithm of Blondel et al. (2008). Louvain greedily maximizes modularity,

$$Q = \frac{1}{2m} \sum_{i,j} \left( A_{ij} - \frac{k_i k_j}{2m} \right) \delta(c_i, c_j), \quad (4.2)$$

where  $m$  is the number of edges,  $A_{ij}$  is the adjacency matrix,  $k_i$  is the degree of node  $i$ , and  $\delta(c_i, c_j)$  indicates whether  $i$  and  $j$  belong to the same community. The method alternates local node moves that improve modularity and graph aggregation steps that

compress communities into super-nodes (Blondel et al., 2008). It is widely used precisely because it scales well on large sparse graphs.

### 4.3 Derivation of the Scalable Complexity

Assume that the graph is partitioned into  $C$  communities of equal size. Then each community has approximately

$$|V_c| = \frac{n}{C} \quad (4.3)$$

vertices. If an exact MIE routine of cost  $O(k^4)$  is applied to a community of size  $k$ , then the total cost across all communities is

$$\sum_{c=1}^C O(|V_c|^4) = C \cdot O\left(\left(\frac{n}{C}\right)^4\right) = O\left(\frac{n^4}{C^3}\right). \quad (4.4)$$

Thus, under the equal-sized-community approximation,

$$T_{\text{scaled}}(n, C) = O\left(\frac{n^4}{C^3}\right). \quad (4.5)$$

Adding community detection itself yields the practical complexity proxy

$$T_{\text{total}}(n, m, C) = O(m) + O\left(\frac{n^4}{C^3}\right). \quad (4.6)$$

The first term corresponds to community decomposition, and the second to independent local exact treatment.

### 4.4 Interpretation of the Approximation

The gain comes from replacing one large exact problem by many smaller exact problems. This is not a free lunch: it assumes that the dominant causes of majority illusion are often local to communities and that inter-community edges can be preserved. This assumption is empirically plausible in modular social networks but still requires validation in simulation, because cross-community ties may transmit cascades even after local neighbourhoods are balanced.

# Chapter 5

## Simulation Setup

### 5.1 Dataset

The implementation uses a public Instagram follower dataset described in the project materials as containing roughly 70 000 users and approximately 1 million edges. The dataset is provided in Appendix 2. The graph is interpreted as a follower network, i.e. a directed graph in which an arc  $(u, v)$  indicates that user  $u$  follows user  $v$ .

### 5.2 Preprocessing Pipeline

The implementation pipeline consists of the following steps:

1. load the follower graph and remove malformed entries, duplicate edges, and isolated artifacts if present;
2. construct an undirected projection for community detection and structural treatment;
3. detect communities with the Louvain method;
4. select communities for exact treatment;
5. reconstruct two comparable directed graphs for simulation: an untreated graph and a treated graph.

### 5.3 Community-Level Treatment and Global Reconstruction

Let  $\mathcal{C}$  be the set of detected communities selected for treatment. The original simulation graph is the induced subgraph on the union of their vertices,

$$G^{\text{orig}} = G[V^*], \quad V^* = \bigcup_{c \in \mathcal{C}} V_c. \quad (5.1)$$

For each community  $c \in \mathcal{C}$ , the exact local MIE solver produces a modified undirected edge set  $E'_c$ . The treated global graph keeps all original cross-community edges among vertices in  $V^*$  and replaces only the intra-community edge sets. This gives

$$G^{\text{treat}} = (V^*, E_{\times} \cup \bigcup_{c \in \mathcal{C}} E'_c). \quad (5.2)$$

This reconstruction is essential for the final experiment, because it isolates the effect of local structural intervention while preserving the original pathways between communities.

### 5.4 LLM Agents

Each node in the simulation graph is mapped to an LLM-based agent. The agent has a continuous opinion state

$$o_v(t) \in [0, 1], \quad (5.3)$$

where larger values correspond to stronger support for the focal minority opinion. A binary label can be derived for structural analysis by thresholding,

$$\ell_v(t) = \mathbb{I}[o_v(t) > 0.5]. \quad (5.4)$$

The initial condition in the reported setting assigns approximately 20% of agents to the minority side.

In each step, an agent may create a post, consume a feed, update its opinion, and adjust its follow graph. The update rule is implemented procedurally through LLM prompting rather than by a closed-form analytical transition function. Nevertheless, conceptually one may view it as

$$o_v(t+1) = \text{LLM}(o_v(t), F_v(t), M_v(t)), \quad (5.5)$$

where  $F_v(t)$  denotes the ranked feed observed by agent  $v$  and  $M_v(t)$  denotes optional memory or history context.

## 5.5 Feed Formation

The feed is rank-based and limited to a maximum of `feed_size=15` items in the baseline experiments. The ranking score is based on open-source data from Instagram software engineers:

$$\text{score}(p) = \text{relevance}(p) + \text{engagement}(p) + \text{recency}(p). \quad (5.6)$$

## 5.6 Simulation Parameters

The core simulation parameters used in the experiment are:

| Parameter                     | Value                    |
|-------------------------------|--------------------------|
| Topic                         | <code>remote work</code> |
| Number of simulated steps $T$ | 20                       |
| Feed size                     | 15                       |
| Initial minority share        | 20%                      |
| Graph size                    | 70023 agents             |
| Edges                         | 1031041                  |

## 5.7 Why LLM Agents Are Used

The rationale for LLM agents is not that they perfectly model humans; it is that they allow a richer response to mixed textual and structural stimuli than a purely threshold-based simulator. This is aligned with the multi-agent simulation perspective of Park et al. (2023). The resulting environment serves as an experimental framework for studying how graph structure, feed composition, and adaptive rewiring jointly affect cascade dynamics.

# Chapter 6

## Experiments and Results

### 6.1 Overview of the Experimental Design

The experimental part evaluates whether community-wise structural treatment based on Majority Illusion Elimination (MIE) can reduce the spread of a negative opinion in a realistic social-network simulation. The experiment is organized around one central comparison: cascade dynamics on the original graph versus cascade dynamics on the treated graph. The implementation code is available in the project repository: Appendix 1.

The full treatment procedure is applied community by community. First, the Instagram follower graph is partitioned into communities. Then, for each selected community, local MIE is solved using the initial binary opinion labels. The resulting edge additions and removals are applied inside the corresponding community. Inter-community edges are not changed. After all communities are processed, the treated graph is assembled from the locally treated communities and the original inter-community structure.

The visualization of Community 18 below is used as an illustrative example of this treatment step. It shows how the same local MIE procedure modifies one community during the construction of the treated graph.

After the treated graph is built, both the original and the treated graphs are initialized with the same initial opinions. In both cases, 20% of agents start with the negative opinion. The LLM-agent cascade simulation is then run separately on the two graphs with the same topic, feed size, and simulation parameters. This design isolates the effect of structural treatment: the only systematic difference between the two runs is the set of intra-community edges changed by MIE.

### 6.2 Empirical Louvain Decomposition

Before applying community-wise MIE, the Instagram graph was decomposed using the Louvain method. The decomposition produced 161 communities in total. The sizes of the

detected communities are highly heterogeneous, which is typical for real social networks: a small number of large communities coexist with many very small groups.

Table 6.1: Louvain community size statistics for the Instagram graph.

| Metric                         | Value  |
|--------------------------------|--------|
| Number of detected communities | 161    |
| Minimum community size         | 3      |
| Median community size          | 13.0   |
| Maximum community size         | 16,545 |
| Mean community size            | 437.32 |

The large gap between the median and the mean indicates that the community-size distribution is strongly skewed. Most communities are very small, while a few communities contain a large fraction of the network.

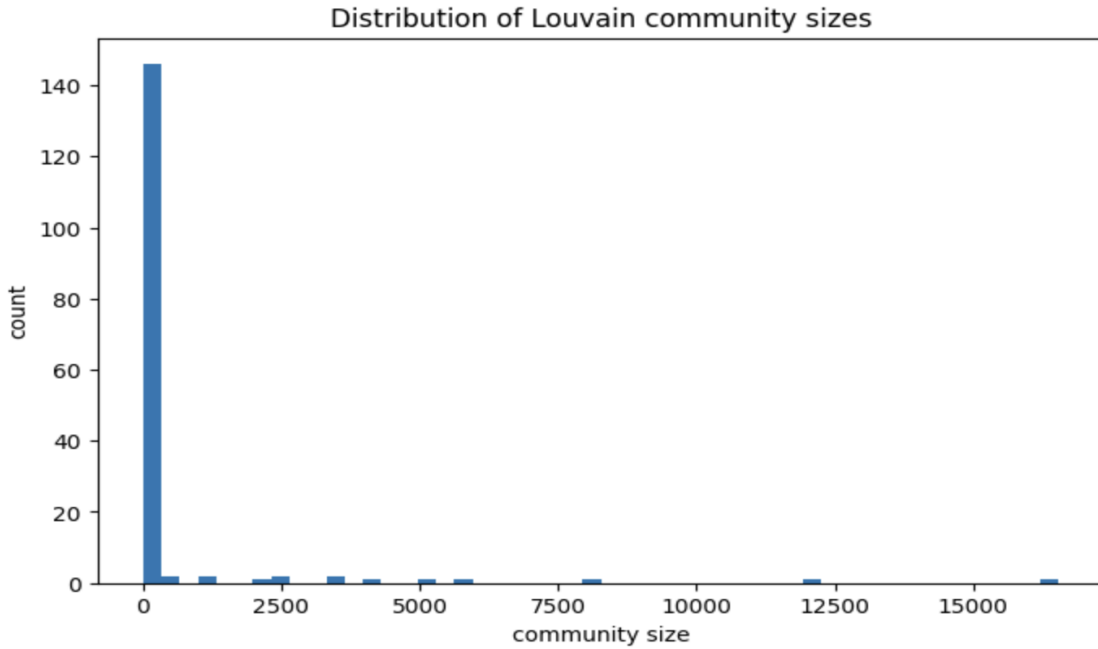


Figure 6.1: Distribution of Louvain community sizes in the Instagram graph.

### 6.3 Community-Level Structural Treatment Example

To make the community-wise treatment procedure explicit, we report one representative example: Community 18. This community contains 272 vertices and 1124 edges before treatment. Under the initial labeling, 19 vertices are under majority illusion. The local MIE solver identifies 12 edge modifications: 10 edge additions and 2 edge removals. After these modifications, the number of illusion nodes in the community becomes zero.

Table 6.2: Illustrative local MIE treatment result for Community 18, used as one step in constructing the global treated graph.

| Metric             | Before treatment | After treatment |
|--------------------|------------------|-----------------|
| Number of vertices | 272              | 272             |
| Number of edges    | 1124             | 1132            |
| Illusion nodes     | 19               | 0               |
| Add candidates     | 23008            |                 |
| Remove candidates  | 54               |                 |
| Objective value    | 12               |                 |
| Edges added        | 10               |                 |
| Edges removed      | 2                |                 |

This example demonstrates the local mechanism used throughout the global treatment stage. The algorithm does not attempt to rewrite the whole graph at once. Instead, it removes local majority illusion inside each community by a small number of edge edits. For Community 18, twelve modifications are enough to eliminate all nineteen illusion nodes, while changing only a small fraction of the community edge structure.

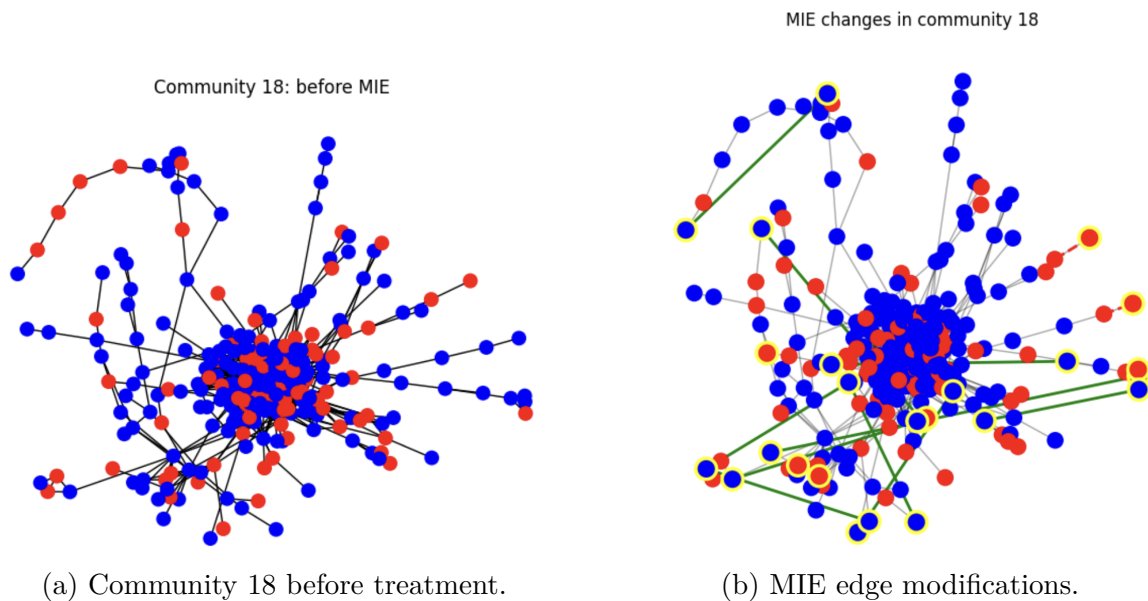


Figure 6.2: Illustration of local MIE treatment for Community 18. Nodes under majority illusion are highlighted. Added edges are marked in green; removed edges are marked with red dashed lines.

## 6.4 Global Original-versus-Treated Experiment

The final experiment evaluates whether local structural treatment changes global cascade dynamics. The procedure is as follows.

First, all vertices belonging to the selected Louvain communities are collected. The original graph is then induced on this vertex set, while keeping all intra-community and inter-community edges from the original Instagram network. Next, each community is treated independently using local MIE. After treatment, the graph is reconstructed: intra-community edges are replaced by treated versions, while inter-community edges are preserved unchanged.

This gives two graphs over the same vertex set:

1. the *original graph*, with all edges exactly as in the source network;
2. the *treated graph*, where intra-community edges are modified by local MIE and inter-community edges remain unchanged.

Both graphs are initialized with the same initial opinions: 20% of agents have the negative opinion. Then the same LLM-agent cascade simulation is run separately on both graphs. The topic used in the simulation is **remote work**. The comparison therefore isolates the effect of structural treatment.

Table 6.3: Global cascade comparison before and after community-wise MIE treatment.

| Metric                         | Original graph | Treated graph |
|--------------------------------|----------------|---------------|
| Number of vertices             | 70,023         | 70,023        |
| Number of edges                | 1,031,041      | 1,032,034     |
| Initial negative-opinion share | 20%            | 20%           |
| Final negative-opinion share   | 58.9%          | 23.1%         |
| Opinion switches               | 27,238         | 2,170         |
| Unfollow events                | 10,284         | 9,883         |
| Follow events                  | 5,987          | 4,839         |
| Posts created                  | 42,101         | 41,891        |
| Community-level edge additions | –              | 1,953         |
| Community-level edge removals  | –              | 960           |
| Net edge change                | –              | +993          |

The difference between the two cascade trajectories is substantial. In the original graph, the negative opinion grows from 20% to 58.9%, which indicates a large cascade. In the treated graph, the final negative-opinion share is stabilized after several steps on

23.1%. This is close to the initial level and suggests that the structural intervention strongly loosens cascade growth.

The number of opinion switches also decreases sharply: from 27,238 in the original graph to 2,170 after treatment. This reduction is important because it shows that the treatment does not merely change the final aggregate state; it changes the dynamic process itself. Agents become less likely to cross the opinion threshold during the simulation.

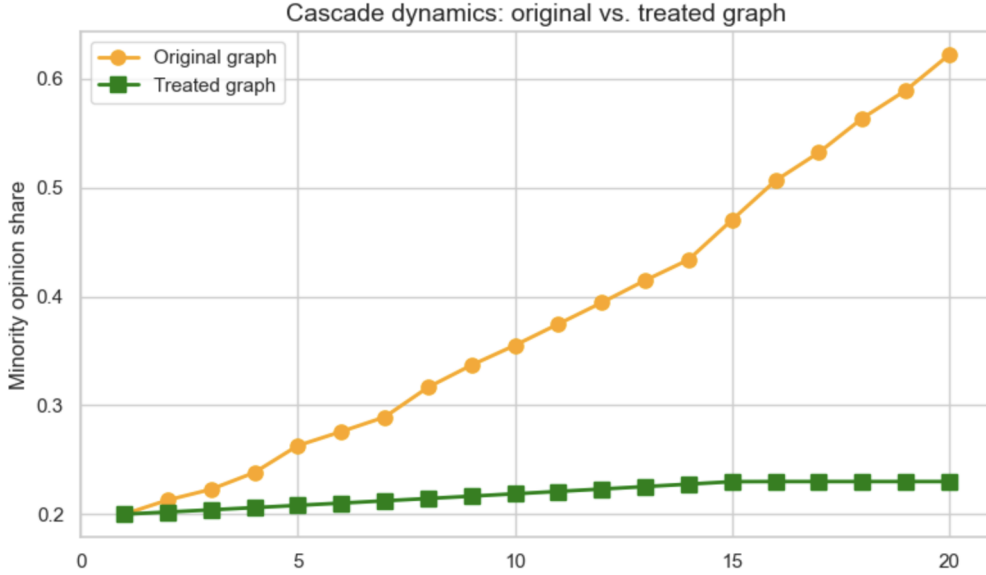


Figure 6.3: Cascade comparison on the original and treated graphs. The plot should show the negative-opinion share over simulation steps for both graphs.

## 6.5 Interpretation of the Results

The community-level and global experiments support the same conclusion: majority illusion can be structurally fragile. At the global level, applying local MIE across communities produces a treated graph whose edge count differs only modestly from the original graph, yet cascade dynamics change dramatically.

The global treatment adds 1,953 edges and removes 960 edges, producing a net increase of 993 edges. Relative to more than one million edges in the graph, this is a small structural perturbation. Nevertheless, the final negative-opinion share drops from 58.9% to 23.1%.

This result is consistent with the main hypothesis of the thesis: local neighbourhood balancing can reduce the conditions under which a minority opinion appears dominant and spreads through the network. The treatment does not need to delete large portions of the graph. Instead, it changes the exposure structure around illusion-prone regions.

# Chapter 7

## Discussion, Conclusion, and Future Work

### 7.1 Discussion of the Results

The experiments show that majority illusion is not only a formal graph-theoretic property, but also a meaningful structural factor in opinion cascade dynamics. The main empirical result is the comparison between the original Instagram graph and the treated graph constructed through community-wise Majority Illusion Elimination. Both graphs contain the same set of 70,023 vertices, both are initialized with the same 20% negative-opinion share, and both are simulated under the same LLM-agent dynamics. The only systematic difference is the structural treatment applied inside communities.

The contrast is large. In the original graph, the negative-opinion share grows from 20% to 58.9%. In the treated graph, the final negative-opinion share is only 23.1%. The number of opinion switches decreases from 27,238 to 2,170. This is perhaps the clearest indication that the treatment changes not merely the final aggregate state, but the underlying cascade process. Fewer agents cross the opinion threshold, and the negative opinion fails to acquire the same self-reinforcing momentum.

A useful way to read this result is through exposure. Majority illusion arises when local neighbourhoods over-represent a globally minority opinion. The treatment does not change opinions directly. Instead, it modifies edges so that local neighbourhoods become less misleading. In the simulation, this structural correction translates into a weaker cascade. The intervention is also relatively small compared to the total graph size: 1,953 edges are added and 960 are removed, while the graph contains more than one million edges. The net change is +993 edges. Yet the cascade outcome changes substantially.

It is also important that post creation remains nearly unchanged in the global experiment: 42,101 posts in the original graph and 41,891 posts in the treated graph. This suggests that the reduction in cascade size is not explained by a major difference

in agent activity. The stronger explanation is structural: agents are exposed to a less illusion-prone neighbourhood composition. Follow and unfollow dynamics are also reduced after treatment, but less dramatically than opinion switches. This is consistent with the idea that structural balancing primarily affects opinion change, while network rewiring remains influenced by other factors, such as feed ranking, engagement, and stochastic agent decisions.

The results support the main hypothesis of the thesis: local structural correction can reduce global cascade risk. At first glance, this may look surprising. How can local changes inside communities affect the whole network? The answer is that cascades often start from repeated local exposure, not from a global broadcast. If communities are treated before the cascade begins, then many local reinforcement paths become weaker. Small changes accumulate.

## 7.2 Limitations

Several limitations should be noted.

First, the MIE treatment is applied locally inside communities, while inter-community edges are preserved unchanged. This choice makes the algorithm scalable and interpretable, but it also means that cross-community illusion is not directly optimized. If a cascade is driven mainly by bridges between communities, then local treatment alone may be insufficient.

Second, the current implementation treats edge modifications as structural changes. In a real social platform, one usually cannot force users to follow or unfollow accounts. A more realistic interpretation is that added edges correspond to increased exposure through recommendations, while removed edges correspond to downranking or reduced exposure in the feed. This connection is conceptually natural, but a production-level feed-ranking formulation would require weighted or probabilistic edges.

Third, the LLM-agent simulation is a proxy for human social behaviour. It allows rich text-based interaction, memory, posting, and opinion updating, but it should not be interpreted as direct evidence about real users. The LLM agents provide a controlled experimental environment. They do not replace empirical behavioural data.

Fourth, computational cost remains a practical issue. Running LLM-based simulations on the entire graph is expensive, because each step may require many API calls. The global experiment demonstrates feasibility at the algorithmic level, but repeated runs, parameter sweeps, and statistical robustness checks require substantial computational resources.

Finally, the model uses binary discretization of opinion through the threshold  $o_v > 0.5$ . This is useful for connecting LLM opinions to majority illusion theory, but real opinions may be multidimensional, uncertain, or topic-dependent.

## 7.3 Conclusion

This thesis developed and tested a scalable framework for detecting and mitigating opinion cascades caused by majority illusion in realistic social network simulations.

The theoretical foundation is the Majority Illusion Elimination problem, where the goal is to modify the network with a minimum number of edge additions and removals so that no user observes a false local majority. Exact MIE algorithms are polynomial, but direct application to large graphs is costly. To address this, the thesis proposed a community-wise scaling strategy: decompose the graph into Louvain communities, apply local MIE inside each community, preserve inter-community edges, and assemble a treated graph.

On the Instagram graph, community-wise treatment changes a small fraction of the edge set but strongly reduces cascade growth. The final negative-opinion share decreases from 58.9% in the original graph to 23.1% in the treated graph, and opinion switches decrease from 27,238 to 2,170.

These findings indicate that majority illusion can be a powerful driver of cascade formation, and that structural treatment of local neighbourhoods can significantly weaken this process. The work also provides a simulation framework in which graph algorithms, feed dynamics, and LLM-based agents can be studied together.

## 7.4 Future Work

Several directions naturally follow from this thesis.

The first direction is to move from unweighted edge editing to weighted exposure optimization. In a real recommender system, treatment would more likely change the probability that a user sees a post, rather than the underlying follower graph itself. A weighted version of MIE could define variables for feed exposure weights and minimize ranking distortion subject to majority-illusion constraints.

The second direction is to include inter-community treatment. The current algorithm preserves edges between communities. A future version could first treat communities locally and then solve a smaller MIE-like problem on the community graph, where each node represents a community and edge weights represent cross-community exposure.

The third direction is broader experimental validation. The current results are promising, but future work should include repeated simulations with different random seeds, different initial minority shares, different topics, and different LLM models. This would make it possible to estimate confidence intervals and test robustness of the treatment effect.

The fourth direction is to extend the opinion model. Binary opinions are useful for theory, but many real discussions involve multiple positions or continuous attitudes.

Extending the model to multi-class or multidimensional opinions would make the simulation more realistic and would connect naturally to generalized  $p$ -illusion formulations.

The fifth direction is fact-checking and content-level intervention. Edge-based treatment modifies exposure structure, while fact-checking modifies the perceived reliability of individual posts. Combining these two mechanisms may produce stronger and more realistic mitigation strategies.

Finally, the framework can be used as a testbed for safer recommender systems. Instead of optimizing only engagement, ranking algorithms could include structural fairness or illusion-reduction terms. This would shift the goal from maximizing attention to maintaining a healthier information environment. That is the broader motivation of the work.

# Bibliography

- Vincent D. Blondel, Jean-Loup Guillaume, Renaud Lambiotte, and Etienne Lefebvre. Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment*, 2008(10):P10008, 2008. doi: 10.1088/1742-5468/2008/10/P10008.
- Damon Centola. The spread of behavior in an online social network experiment. *Science*, 329(5996):1194–1197, 2010. doi: 10.1126/science.1185231.
- Jack Dippel, Max Dupré la Tour, April Niu, Sanjukta Roy, and Adrian Vetta. Eliminating majority illusion is easy. In *Proceedings of the Thirty-Ninth AAAI Conference on Artificial Intelligence*, pages 13763–13770, 2025. Extended version appeared on arXiv as arXiv:2407.20187.
- W. Dvořák et al. Eliminating majority illusions in networks, 2023. Bibliographic metadata were unspecified in the provided project materials; this placeholder entry is retained only to mirror the requested bibliography and should be replaced with the exact record before final submission if department policy requires full verification.
- Umberto Grandi, Lawqueen Kanesh, Grzegorz Lisowski, M. S. Ramanujan, and Paolo Turrini. Identifying and eliminating majority illusion in social networks. In *Proceedings of the Thirty-Seventh AAAI Conference on Artificial Intelligence*, volume 37, pages 5062–5069, 2023. doi: 10.1609/aaai.v37i4.25634.
- Mark Granovetter. Threshold models of collective behavior. *American Journal of Sociology*, 83(6):1420–1443, 1978. doi: 10.1086/226707.
- Matthew O. Jackson. *Social and Economic Networks*. Princeton University Press, Princeton, NJ, 2010.
- Kristina Lerman, Xiaoran Yan, and Xin-Zeng Wu. The “majority illusion” in social networks. *PLOS ONE*, 11(2):e0147617, 2016. doi: 10.1371/journal.pone.0147617.
- Joon Sung Park, Joseph C. O’Brien, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*, pages 1–22, 2023. doi: 10.1145/3586183.3606763.

# Appendix A

## Code and Reproducibility

1. The source code for the experiments is available at:

[https://github.com/AkbarIsakov/  
Treating-the-Majority-Illusion-cascades-in-LLM-based-Instagram-graph-simulation](https://github.com/AkbarIsakov/Treating-the-Majority-Illusion-cascades-in-LLM-based-Instagram-graph-simulation)

2. The experiments use the public Kaggle dataset *Instagram 70K*.

<https://www.kaggle.com/datasets/krpurba/im-instagram-70k>